A Quadratic WSPD Obstruction in Nagata Dimension Zero

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Source Question

The source paper is Charles Fefferman and Pavel Shvartsman, *Sharp finiteness principles for Lipschitz selections: long version*, arXiv:1708.00811. In the introduction, after explaining that the paper's results lead to efficient-computation questions for Lipschitz selection problems on finite metric spaces, the authors ask whether the results of Har-Peled and Mendel on well-separated pair decompositions for doubling metrics can be extended from doubling metrics to metrics of bounded Nagata dimension.

This packet gives a negative answer to the direct linear-size WSPD extension: bounded Nagata dimension alone cannot imply an $O(n)$-size WSPD for fixed separation.

Definitions

We use the Nagata condition as stated in the source. A metric space (X, d) satisfies the Nagata condition with constants $c \in (0, 1]$ and $C \geq 0$ if for every $r > 0$ there is a cover of X by subsets of diameter at most r, at most $C + 1$ of which meet any ball of radius cr.

For a finite metric space X and a parameter $s > 0$, an s-well-separated pair decomposition, abbreviated s-WSPD, is a family of pairs (A_i, B_i) of nonempty disjoint subsets of X such that every unordered two-point set $\{x, y\} \subset X$ is covered by at least one pair, meaning $x \in A_i, y \in B_i$ or $y \in A_i, x \in B_i$, and

$$\operatorname{dist}(A_i, B_i) \geq s \max\{\operatorname{diam} A_i, \operatorname{diam} B_i\}.$$

This is the standard metric form of the WSPD separation condition. If one's convention asks for each two-point set to be covered exactly once, the lower bound below remains valid.

Proof Intuition

Bounded Nagata dimension controls covers scale by scale, but it does not control packing. The equilateral metric is the extreme example: its Nagata dimension is 0, because below the unit scale singletons work and at or above the unit scale the whole space works. But a WSPD with separation $s > 1$ cannot group two equilateral points on either side of a pair: a set with two points already has the same diameter as its distance to any disjoint set. Therefore every WSPD pair must be singleton versus singleton, so all $\binom{n}{2}$ point pairs must be listed separately.

Lemma 1. *For each $n \geq 2$, the n-point equilateral metric space E_n, with $d(x, y) = 1$ for $x \neq y$, satisfies the Nagata condition with constants $c = 1$ and $C = 0$.*

Proof. Fix $r > 0$. If $r < 1$, cover E_n by its singleton subsets. Each has diameter 0, and each ball of radius r contains only its center, hence meets exactly one member of the cover. If $r \geq 1$, cover E_n by the single set E_n, whose diameter is $1 \leq r$. Again every ball meets only one member of the cover. This proves the Nagata condition with $c = 1$ and $C = 0$, uniformly in n. $\square$

Lemma 2. *Let $s > 1$. Every s-well-separated pair (A, B) in E_n has $\#A = \#B = 1$.*

Proof. If A contains two distinct points, then $\operatorname{diam} A = 1$. Since A and B are nonempty and disjoint, $\operatorname{dist}(A, B) = 1$. The s-separation condition would require $1 \geq s$, a contradiction. Hence A is a singleton. The same argument applies to B. $\square$

Theorem 1. *For every $s > 1$ and every $n \geq 2$, every s-WSPD of the n-point equilateral metric E_n has at least $\binom{n}{2}$ pairs. Consequently, there is no bound of the form $C(s, c, C)\, n$ for WSPDs on all finite metric spaces satisfying the Nagata condition with constants c, C.*

Proof. By the preceding lemma, each s-well-separated pair in E_n is of the form $(\{x\}, \{y\})$ with $x \neq y$. Such a pair covers exactly the unordered two-point set $\{x, y\}$. Since a WSPD must cover all $\binom{n}{2}$ unordered two-point subsets of E_n, it must contain at least $\binom{n}{2}$ pairs.

The Nagata constants of E_n are $c = 1$, $C = 0$, uniformly in n. Thus no WSPD theorem depending only on s, c, C can give a linear-size bound $C(s, c, C)n$ for all finite bounded-Nagata-dimension metrics. $\square$

Scope and Limitations

The theorem refutes the direct transfer of the usual fixed-separation, linear-size WSPD conclusion from doubling metrics to bounded Nagata dimension. It does not rule out weaker decompositions, randomized or approximate surrogates, or algorithms for Lipschitz selection that exploit bounded Nagata dimension without producing a standard WSPD of linear size.

Novelty Check

The run indexes were searched for arXiv:1708.00811, the source title, 'Lipschitz selections', 'finiteness principle', 'WSPD', 'Har-Peled Mendel', 'doubling metrics', and 'Nagata dimension'. No exact prior packet was found. A bounded web search on 2026-07-19 for the exact WSPD/Nagata phrases found the source paper and general Nagata references, but no exact equilateral lower bound recorded as an answer to this source question.

References

References

[1] C. Fefferman and P. Shvartsman, *Sharp finiteness principles for Lipschitz selections: long version*, arXiv:1708.00811.

[2] P. B. Callahan and S. R. Kosaraju, *A decomposition of multidimensional point sets with applications to k-nearest-neighbors and n-body potential fields*, J. ACM 42 (1995), 67–90.

[3] S. Har-Peled and M. Mendel, *Fast construction of nets in low-dimensional metrics and their applications*, SIAM J. Comput. 35 (2006), no. 5, 1148–1184.